

Lab - 2 - Python Basics (Conditional Statements, Tuples, Dictionaries and Arrays)

1. _____ Implement if, if...else, and elif statements, Practice loops (for, while) with examples.
2. _____ Complete the following tasks using Tuples -
 - I. _____ Tuple Creation and Deletion
 - II. _____ Tuple Duplication
 - III. _____ Tuple Length
 - IV. _____ Accessing Tuples - Positive and Negative Indexing, Range of Positive & Negative Indexes, Checking
 - V. _____ Tuple Conversion to a List
 - VI. _____ Adding and Removing Items to a Tuple
 - VII. _____ Looping through Tuples & Tuple Items - for & while
 - VIII. _____ Tuple Joining & Multiplication
 - IX. _____ Using count () & index() method in tuples
3. _____ Complete the following tasks using Arrays -
 - I. _____ Array Creation
 - II. _____ Array Element Access
 - III. _____ Array Length
 - IV. _____ Looping Array Elements
 - V. _____ Adding Array Elements using append()
 - VI. _____ Removing array elements using pop()
4. _____ Complete the following tasks using Sets -
 - I. _____ Set Creation and Set Length
 - II. _____ Allocation of Different Data Types into a Set
 - III. _____ Using set() constructor
 - IV. _____ Accessing Set Items
 - V. _____ Adding items to a Set using add() and update()
 - VI. _____ Remove set item using remove (), pop() and discard()
 - VII. _____ Use clear() and del() on a set
 - VIII. _____ Set Looping - for and while
 - IX. _____ Set Joining using union(), update(), intersection(), intersection_update()
 - X. _____ Use difference(), difference_update(), isdisjoint() and symmetric_difference()
5. _____ Complete the following tasks using Dictionaries -
 - I. _____ Dictionary Creation & Access using Key, and Updation
 - II. _____ Dictionary Length, Type and dict() constructor
 - III. _____ Dictionary Adding and Removing Items (pop(), popitem()) and Checking
 - IV. _____ Using clear() and del() on Dictionary
 - V. _____ Looping and Copying Dictionary using copy()
 - VI. _____ Nested Dictionary - Creation, Accessing and Looping Items